



Advanced Short Circuit Preventive Mechanism for Electric Vehicles

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Objective

By creating a cutting-edge short circuit prevention system, our initiative seeks to transform electrical safety in electric vehicles. The system improves safety and guarantees proactive risk reduction by utilising self-healing technology, AI-based predictive analytics, and real-time monitoring. With an emphasis on system dependability, energy efficiency, and scalability, this solution is made to safeguard vital parts, minimise downtime, and easily adjust to new developments in technology, establishing a new benchmark for safe and sustainable electric vehicle operations.

Literature Review Table

Author	Title	Year of published	Drawback
D. Qiao et al	"Quantitative Diagnosis of Internal Short Circuit for Lithium-Ion Batteries Using Relaxation Voltage"	2024, IEEE Transactions on Industrial Electronics	Focuses primarily on diagnosis without providing real-time mitigation strategies for detected faults.
Z. Sun et al.	"An Online Data-Driven Fault Diagnosis and Thermal Runaway Early Warning for Electric Vehicle Batteries"	2022, IEEE Transactions on Power Electronics	Limited to battery systems; lacks consideration of external factors like environmental conditions.
F. Kaya and O. Akar	"Short Circuit Effects on HV Feeders of Optimally Located Electric Vehicle Fast Charging Stations"	2024, IEEE Access	Addresses stationary systems only; does not factor in dynamic or mobile EV charging scenarios.

Existing System

- ☐ Fuse Protection
- ☐ Current Limiting Device
- ☐ Electronic Safety Relays
- ☐ Ground Fault Detection

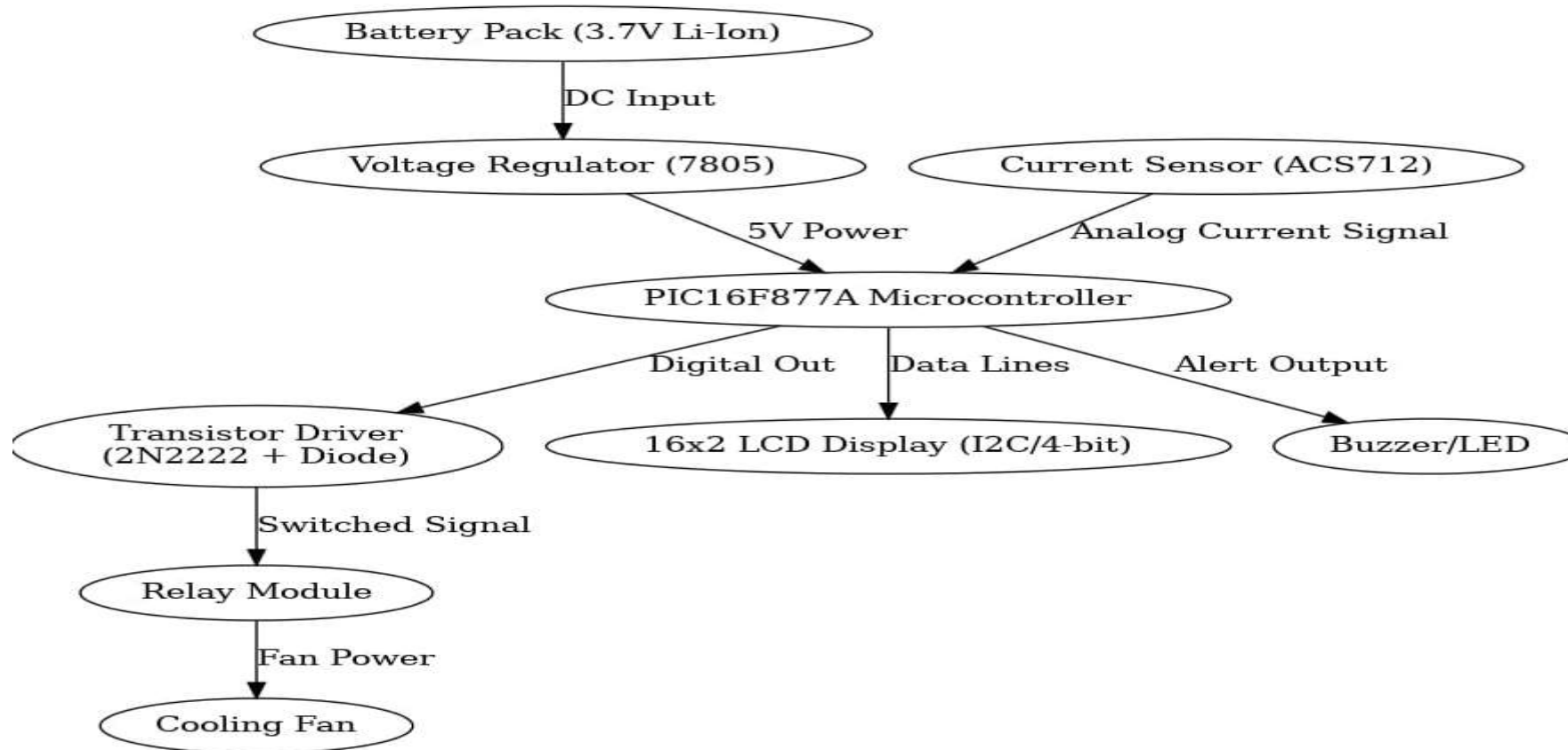
Proposed Methodology

- ☐ Cell Imbalancing detection
- ☐ AI-Based Predictive Analytics
- ☐ Automated Circuit Isolation Mechanism
- ☐ Real-time Voltage and Current Monitoring

Plan for Implementation

- ☐ Collect real-time data such as current, voltage, temperature, and cell imbalance using onboard sensors.
- ☐ Extract meaningful features like voltage deviation, temperature rise rate, and current spikes. Normalize data for consistency.
- ☐ Long Short-Term Memory (LSTM) networks.
- ☐ When LSTM predicts a risk, a microcontroller instantly activates a relay to isolate the battery and prevent damage.

Block Diagram



Working of Proposed System

- ☐ Sensors monitor real-time voltage, current, temperature, and cell imbalance, sending data to a PIC microcontroller for local processing.
- ☐ The data is simultaneously updated to the cloud for remote monitoring and further analysis.
- ☐ An LSTM-based machine learning model runs on edge/cloud to predict short circuit risks and trigger a relay-based cut-off if needed.
- ☐ An efficient cooling system using liquid cooling with phase-change material (PCM) or active air-cooled fins maintains optimal battery temperature during high-load conditions.

Importance of ML Algorithm

- ☐ Early Detection of Short Circuits
- ☐ Pattern-Based Analysis
- ☐ Adaptive Learning System
- ☐ Enhanced Battery Safety

Components Used

- ☐ PIC16F877A PIC Microcontroller
- ☐ ESP8266 Wi-Fi Module
- ☐ Current Sensors (ACS712)
- ☐ Cell Voltage Monitor (ADS1115)
- ☐ Temperature Sensor(DS18B20)
- ☐ Two cylindrical Li-ion cells
- ☐ LED Indicators
- ☐ Relay Module
- ☐ Active Cooling System

Expected Results

- ☐ Automated Circuit Isolation
- ☐ Systematic Component Disconnection
- ☐ Fault Prediction by continuously Monitoring the electrical Parameters
- ☐ Real-time Alert

Advantages

- ❑ Real-time detection allows quicker response to short circuits, minimizing damage.
- ❑ AI-based predictive analytics enhance maintenance scheduling, reducing downtime.
- ❑ Automated circuit isolation ensures rapid protection
- ❑ System recovery without manual intervention.

Applications

- ☐ Electric Vehicle Safety Systems
- ☐ Autonomous & Connected Vehicles
- ☐ Battery Energy Storage Systems (BESS)
- ☐ Smart Grids & Power Distribution

Project Outcome Status as on date Detail(Proof)





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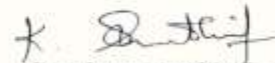
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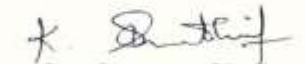
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Reference

- [1] D. Qiao et al., "Quantitative Diagnosis of Internal Short Circuit for Lithium-Ion Batteries Using Relaxation Voltage," in IEEE Transactions on Industrial Electronics, vol. 71, no. 10, pp. 13201-13210, Oct. 2024, doi: 10.1109/TIE.2023.3342289.
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